- **Reed:** A vertical wooden frame with very close steel wires. Each warp yarns pass through each wire. The gap between the wires is known as dents (Figure 5.6).



Figure 5.6 Reed

- **Breast Beam:** This is a bar, which guides the newly woven cloth towards the cloth beam.
- **Cloth Beam:** A wood or steel roller on to which the woven cloth is wound after weaving (Figure 5.7).



Figure 5.7 Cloth Beam



Who discovered the first automatic loom?



Jacques de Vaucanson a French artist who lived between February 24, 1709 – November 21, 1782 was the first man to design an automatic loom.

5.3.1 Basic Weaving Operation

The yarns intended for warp must pass through some basic operations like– spooling, and warping to prepare them to withstand the strain during weaving.

5.3.1.1 Spooling

Winding of warp yarns on large spools, or cones which are placed on a rack called creel is known as spooling. These yarns are then wound on the warp beam (Figure 5.8).



Figure 5.8 Spooling

5.3.1.2 Sizing

The warp yarns are unwound and dipped in a sizing bath which consists of starch. The slasher machine covers every yarn with a coating which adds strength to the warp yarns to withstand the stress, during weaving (Figure 5.9).

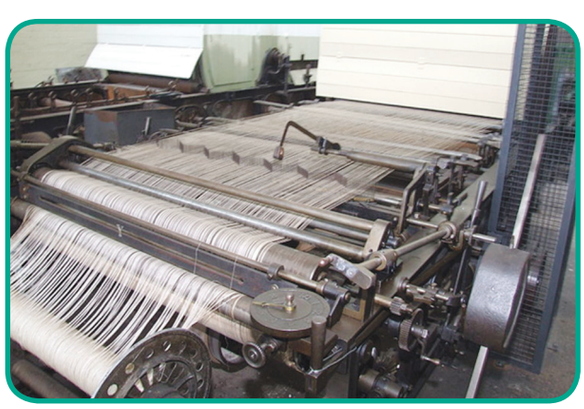


Figure 5.9 Sizing

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